VEDANTU TELUGU JEE CHANNEL

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MATHEMATICS – IIB [75 MARKS]

SECTION - A

I. Answer the following questions ***10x2=20M***

1. If length of tangent from (2, 5) to the circle $x^2+y^2-5x+4y+k=0$ is $\sqrt{37}$ then find "K".
2. If the circle $x^2 + y^2 + ax + by - 12 = 0$ has the centre at (2, 3) then find a, b and the radius of the circle
3. If (4, 2), (K, -3) are conjugate points w.r.t the circle $x^2+y^2-5x+8y+6=0$ then find 'K'
4. Find the equation of the parabola whose vertex is (3,-2) and the focus is (3, 1)
5. If e, e_1 are the eccentricities of a hyperbola and its conjugate hyperbola, prove that $\frac{1}{e^2}+\frac{1}{e_1^2}=1$
6. Evaluate $\int e^x \frac{(1+x\log x)}{x}dx$
7. Evaluate $\int \frac{dx}{x\log x\log(\log x)}$
8. Evaluate $\int_0^{\pi}\sqrt{2+2\cos\theta}.d\theta$
9. Evaluate $\int_0^{\pi/2}\sin^6 x\cos^4 x\,dx$
10. Find the order and degree of $\left[\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^3\right]^{\frac{6}{5}}=6y$

SECTION –B

II. Answer the following ANY FIVE questions ***5x4=20 M***

11. Find the equation of the circle which touches the circle $x^2+y^2-2x-4y-20=0$ externally at (5, 5) with radius 5 units
12. If two circles $x^2+y^2+2gx+2fy=0$ and $x^2+y^2+2g^1x+2f^1y=0$ touch each other then show that $f^1g=fg^1$

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13. Find the length of major axis , minor axis, latus rectum, eccentricity, coordinates of centre, foci and the equations of directrices of the ellipse $9x^2+16y^2=144$

14. Find the condition for the line $x\cos\alpha + y\sin\alpha = p$ to be a tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

15. Find the centre, eccentricity, foci, length of latus rectum and equations of the directrices of the hyperbola $9x^2-16y^2+72x-32y-16=0$

16. Find $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}\,dx$

17. Solve $(x^2+y^2)dx = 2xy\ dy$

SECTION - C

III. Answer the following ANY FIVE questions **5x7=35M**

18. If (2, 0) (0,1)(4,5) and (0,C) are concyclic then find C.

19. Find the equations of the pair of direct common tangents to the circles $x^2+y^2+22x-4y-100=0$ and $x^2+y^2-22x+4y+100=0$

20. Derive Equation of the parabola in standard form with diagram

21. Find $\int \frac{2Sinx+3Cosx+4}{3Sinx+4Cosx+5}dx$

22. Find the reduction formula for $\int \sin^n x$ dx hence find $\int \sin^4 x$ dx

23. Show that $\int_0^1 \frac{\log(1+x)}{1+x^2}dx = \frac{\pi}{8}\log 2$

24. Solve $\frac{dy}{dx}+\frac{y^2+y+1}{x^2+x+1}=0$

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